

Yousif Shasha

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Summary

Structural Engineering student experienced in **water systems analysis**, **field surveying**, **technical drawings**, and **materials testing**. Seeking to bring proven **organization**, **attention to detail**, **team coordination**, **communication**, and **problem solving** to Hatch's Winter 2027 water infrastructure construction co-op.

Education

Toronto Metropolitan University

B.Eng. Structural Engineering

Toronto, ON

Expected 2029

- Relevant Coursework: Project Management, Surveying, Fluid Mechanics and Hydraulics, Concrete Materials, Foundation and soils Engineering, Civil Engineering Systems, Structural Design and Analysis, Strength of Materials, Computer-Aided Structural Analysis, Structural Concrete and Steel Design, Geology for Engineers, and Transportation Engineering.
- Awards: *Dean's List - Fall 2025, Winter 2026. Top 3 Project, Geology for Engineers*

Experience

Groupe Adonis

Grocery Supervisor

Mississauga, ON

July 2023 – Present

- Supervise 2-6 employees, coordinate daily tasks, and train new hires to maintain efficient workflows and teamwork.
- Manage inventory, stocking, palletizing, and organization of 10,000+ products to ensure availability and accurate placement.
- Problem-solve customer and operational issues using leadership, communication, and adaptability in a fast-paced setting.

Projects

Hydraulics Engineering — *EPANET, HEC-RAS, Excel*

Winter 2026

- Investigated pipe losses, pressurized flow, weirs, and sluice gates in seven laboratory experiments.
- Modelled roughness, diameter, demand, and redundancy effects on water networks in EPANET.
- Compared flood profiles through the 100-year event in HEC-RAS and Excel to assess velocity and erosion risk.

Topographic Surveying — *Total Station, Excel*

October 2025

- Led a six-person field team, coordinating measurement sequences and consistent observation procedures.
- Processed total-station readings and elevation calculations with a 0.0032 m benchmark misclosure.
- Organized field data in Excel, checked survey closure, and adjusted traverse coordinates.

Structural Mechanics — *AutoCAD, Excel*

Fall 2025 - Winter 2026

- Prepared AutoCAD drawings showing specimen geometry, dimensions, supports, loads, and gauge locations.
- Calculated stress, deflection, truss forces, and torsion by hand and in Excel.
- Checked calculations against gauge readings, finding a 1.14% neutral-axis difference.

Geotechnical Engineering Laboratory

Winter 2026

- Classified soils using Atterberg limits and sieve analysis to assess plasticity and gradation.
- Evaluated compaction, permeability, and consolidation; measured a maximum dry unit weight of 19.54 kN/m³.
- Assessed soil strength through direct shear and compression tests, measuring a 37.4° friction angle.

Leadership & Activities

Toronto Metropolitan University Concrete Canoe Team

Construction Team Member

Toronto, ON

September 2025 – Present

- Cut, sanded, taped, and assembled canoe components to meet competition requirements.
- Tested materials and fabrication methods to improve structural performance and constructability.
- Documented prototype results and shared findings with teammates to guide design improvements.

Skills

Technical: AutoCAD, Microsoft Excel, EPANET, HEC-RAS, Revit, ETABS (Basic), MATLAB, LaTeX, Python

Engineering: Surveying, Hydraulics, Materials Testing, Technical Reporting, Data Analysis.

Soft: Team Coordination, Communication, Organization, Attention to Detail, Problem Solving, Adaptability.